

Towards global quantum key distribution

Haoran Zhang^{1,2,6}, Haotao Zhu^{1,2,6}, Ruihua He², Yan Zhang², Chao Ding², Lajos Hanzo³✉ & Weibo Gao^{1,2,4,5}✉

Abstract

Quantum key distribution (QKD) is a cryptographic technology that supports the negotiation and sharing of private keys with unconditional security between authorized parties. As QKD scales to a global level, it must address performance limitations, high costs and practical security concerns. In this Review, we outline the key technical challenges, applications and prospective developments towards a global QKD network. Advances such as satellite-based QKD and newly developed protocols offer promising solutions for extending QKD over long distances. Field trials have progressively expanded from intercity links to larger-scale networks. Nevertheless, balancing cost–performance and security considerations will continue to challenge advanced research efforts. On the basis of the strategies addressing these obstacles, we highlight future directions that can support the efficient realization of global QKD infrastructures.

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¹School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore, Singapore.

²Division of Physics and Applied Physics, School of Physical and Mathematical Sciences, Nanyang Technological University, Singapore, Singapore. ³School of Electronics and Computer Science, University of Southampton, Southampton, UK. ⁴Centre for Quantum Technologies, Nanyang Technological University, Singapore, Singapore. ⁵Quantum Science and Engineering Centre (QSec), College of Engineering, Nanyang Technological University, Singapore, Singapore. ⁶These authors contributed equally: Haoran Zhang, Haotao Zhu.

✉e-mail: hanzo@soton.ac.uk; wbgao@ntu.edu.sg

Introduction

In an information-driven society, concerns about information security are paramount, where authorized users face potential adversaries. This ongoing conflict, driven by complementary concerns about protecting information versus the desire to intercept it, can be quantified by comparing the computational resources and information access capabilities of both authorized users and adversaries. Encryption remains the most common method for secure communication; however, the security of most cryptographic schemes relies on the limited computational power of adversaries, thus providing computational security rather than information-theoretical security. Quantum communication imposes natural limitations on an adversary's ability to access information by engaging quantum mechanical properties¹. In quantum cryptography, quantum key distribution (QKD)² garners interest owing to the intuitive role of securely shared keys in encryption. QKD enables the negotiation and sharing of private keys between distant users at information-theoretical security and is composable³ with other cryptographic schemes. Thus, QKD provides a versatile option for encryption in communication schemes that require a pre-shared private key.

Classical networks, such as undersea fibre-optic cables, terrestrial fibre-optic networks and satellite-based systems, are already in use worldwide, enabling seamless communication between users. Therefore, a global QKD network is in demand to protect the overall security of classical networks. Although not all communications may require quantum-secure encryption, it is crucial to ensure that the backbone of the network is covered. However, QKD faces questions regarding its practical value and reliability. The primary challenge comes from post-quantum cryptography (PQC)⁴, which claims to be secure against existing quantum algorithms. Although QKD maintains its unique advantage by offering information-theoretical security, PQC can still alleviate public concerns about the threat posed by quantum computers⁵ to information security. Another concern is QKD's inherent vulnerability to denial-of-service attacks. Unlike classical denial-of-service attacks, those targeting QKD networks occur at the physical layer, which is easier to monitor⁶. However, these concerns have already negatively impacted the reputation of reliability of QKD. Thus, the QKD community must reassure the public⁷ before global adoption.

As QKD technology advances, its various components reach maturity at vastly different rates. For example, some aspects such as Bennett–Brassard 1984 (BB84)-like protocols² have moved beyond scientific research and are being explored for commercial applications. By contrast, the envisioned backbone for all QKD applications, the quantum memory network⁸, remains in its early stages. Consequently, predicting the optimal design of a QKD network is premature^{9,10}. Nevertheless, it is possible to examine QKD research and anticipate future developments with greater clarity.

In this Review, we introduce the fundamentals of QKD implementation and discuss the main challenges confronting its applications. We further explore several approaches for achieving long-distance QKD and discuss the advantages and drawbacks of these methods through their applications. Chiefly, efforts aiming to extend the range of QKD are foundational for building a global quantum communication network. Finally, we outline the developments of both terrestrial and satellite-based QKD networks. We serve as both an introduction and an inspiration for fostering collaboration among academia, government and industry to build a global QKD network.

Realizing QKD networks

The implementation of QKD first requires a protocol designed for the preparation, transmission and measurement of quantum states, followed by post-processing to negotiate the key. Compared with classical communication, these procedures demand additional quantum devices and channels to handle QKD protocols.

QKD protocols

QKD protocols are primarily categorized into two types: discrete-variable (DV) and continuous-variable (CV) protocols (Fig. 1). In DV-QKD, information is mapped, for example, onto single photons or weak coherent pulses² in a discrete manner. These photons are subsequently detected using single-photon detectors¹¹. DV-QKD protocols utilize quantum systems that are described by finite-dimensional Hilbert spaces, including states of polarization², phase¹¹ or orbital angular momentum¹². Therefore, protocols such as differential-phase-shift¹³, coherent-one-way¹⁴ and round-robin differential-phase-shift¹⁵ are also recognized as DV protocols. These distributed-phase-reference protocols are sometimes listed separately owing to the unique challenges they present in security analysis^{16,17}.

Conversely, CV-QKD encodes information onto the field quadrature amplitudes of quantized electromagnetic fields, which are detected using homodyne¹⁸ or heterodyne¹⁹ detectors. The primary CV-QKD protocols use coherent²⁰ or squeezed²¹ Gaussian states encoded with Gaussian or discrete keys within infinite-dimensional Hilbert spaces. Additionally, protocols that utilize thermal states²², discrete modulation²³ or unidimensional preparation²⁴ are also classified as CV protocols.

Although the goal of both DV and CV protocols is the distribution of secret keys, differences in modulation and measurement affect the device requirements and performance in specific scenarios. Gaining an understanding of the various protocols provides a comprehensive perspective for implementing QKD networks.

Quantum devices for QKD

The implementation of QKD necessitates the use of specialized quantum devices, primarily consisting of a light source, modulators and detectors. Many of these devices are similar to those used in classical optical communications (Fig. 1). For the light source, a single-photon source is ideal for many DV-QKD protocols. Weak coherent pulses associated with the decoy-state method^{25,26} are more commonly used as light sources to simulate single-photon sources owing to their robustness and efficiency. For CV-QKD, homodyne or heterodyne detectors, in which the signal interferes with a local oscillator, are required. For DV-QKD, the most indispensable devices are single-photon detectors, such as single-photon avalanche diodes (SPADs)²⁷ and superconducting nanowire single-photon detectors (SNSPDs)²⁸. Single-photon detectors can precisely provide both the presence and timing of a single photon arriving at the detectors. Assisted by the optical design at the receiver, these detectors effectively perform the positive operator-valued measurements²⁹.

For near-infrared SPADs, the avalanche photodiode typically uses an indium gallium arsenide (InGaAs) layer for absorption and either an indium phosphide (InP) or indium aluminium arsenide (InAlAs) layer for multiplication³⁰. SPADs work in Geiger mode³¹, in which the absorption of a single photon can initiate the impact-ionization process. Therefore, the emerged macroscopic, self-sustaining avalanche current is detectable³¹. However, the electrical avalanche amplification structure is highly sensitive to environmental noise, and even noise

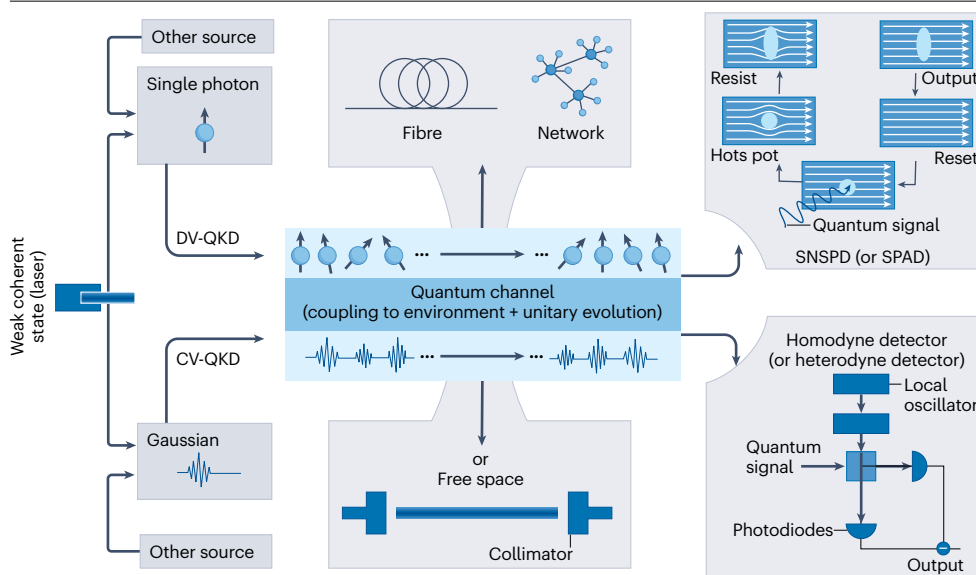


Fig. 1 | Schematic of quantum key distribution systems using discrete-variable and continuous-variable encoding. From an implementation perspective, discrete-variable (DV) and continuous-variable (CV) schemes are largely similar, as they can share the same sources and channels. The weak coherent state is used to simulate a single-photon source in DV-quantum key distribution (QKD), as its Poisson distribution primarily consists of vacuum and single-photon components. The quantum channel is commonly implemented via fibre or free-space links. DV-QKD typically uses superconducting nanowire single-photon detector (SNSPD) or single-photon avalanche diode (SPAD), whereas CV-QKD relies on homodyne or heterodyne detection.

from the avalanche process itself, resulting in a higher dark count rate and an increased probability of afterpulsing³². Thus, a quenching circuit is essential to suppress the avalanche and reset the SPAD to its initial bias condition. This process introduces additional dead time as a trade-off. Optimizing the design of the avalanche photodiode structure, such as increasing the multiplication region thickness³³, can reduce the dark count rate but further compromises photon detection efficiency (PDE).

The performance of SNSPD surpasses that of SPAD across most metrics. SNSPDs have a shorter reset time for returning to the superconducting state after detection and demonstrate superior PDE and dark count rate performance, particularly in the near-infrared spectrum. For example, the typical PDE of InGaAs SPAD is around 30%, whereas SNSPD can achieve a maximum PDE of up to 99.5% (ref. 34). Similarly, although an SPAD typically exhibits a dark count rate around several tens of thousands of Hertz, SNSPDs can reduce this to as low as 0.1 Hz (ref. 35).

The operation of SNSPD is based on a mechanism in which a supercurrent assists in the formation of non-superconducting regions, enhancing the PDE, time precision and noise robustness. Typically, the nanowire is made of materials such as niobium nitride (NbN)³⁶, niobium titanium nitride (NbTiN)³⁴, tungsten silicide (WSi)³⁷ or molybdenum silicide (MoSi)³⁵ and is cooled to well below its superconducting transition temperature and biased with a constant current near its critical value. In this superconducting state, Cooper pairs – charge carriers formed through electron–phonon interactions – are stable under Bose–Einstein condensation³⁸. Upon photon absorption, if the energy of the photon surpasses the binding energy of a Cooper pair, it breaks into two quasiparticles. For example, a 1,550 nm photon can break 125 Cooper pairs in an NbN superconductor³⁸. The generation of hundreds of quasi-particles forms a non-superconducting ‘hot spot’. Subsequently, increased current density and vortex-assisted mechanisms³⁹ cause the hot spot to expand, eventually encompassing the entire cross-section of the nanowire. This expansion leads to an instantaneous increase in resistance, which is then detected.

Quantum channel

Besides authenticated classical channels, quantum channels are required for sharing quantum resources between authorized users in the general settings of QKD protocols. Quantum channels establish a physical connection between QKD users, allowing quantum states to be transmitted at the lowest possible environmental perturbation. Optical fibres and free-space links are the most commonly used quantum channels (Fig. 1). Single-mode fibres are the preferred choice, as telecommunications networks are already extensively built on single-mode fibres operating at telecommunication wavelengths. Because single photons cannot be amplified like classical signals, the low-loss and stable transmission characteristics of telecom-band single-mode fibres make them ideal for quantum state transmission. Although issues such as birefringence, scattering⁴⁰, group-velocity dispersion and polarization mode dispersion⁴¹ still exist, even in underground dark fibres⁴², their effects can be substantially mitigated by noise-reducing strategies²⁹. As a result, a QKD field test over 500 km has been successfully achieved⁴³. The establishment of a free-space quantum channel offers greater flexibility than fibre optics, especially when navigating challenging terrains or setting up temporary links. Additionally, free-space channels can cover much longer distances than fibre channels as the attenuation in free space is less than that in fibre. For instance, the satellite-to-ground channel attenuation varied from 29 dB at 530 km to 44 dB at 1,600 km in Micius experiment⁴⁴. However, additional noise factors such as stray light and turbulence need to be addressed. Stray light can cause serious background noise, whereas turbulence can lead to space-time redistribution as well as beam spreading and wandering, causing a higher error rate of quantum states. Consequently, the use of spectral, spatial and temporal filters, along with careful selection of operating times, spaces, wavelengths and quantum basis, is an essential consideration to avoid or compensate for noise^{45,46}. For an unmanned aerial vehicle⁴⁷ or satellite QKD⁴⁸, developing a fast optical tracking system is also crucial for maintaining a stable and low-loss quantum channel⁴⁹. Furthermore, the quantum channel for terahertz QKD can be readily established using wireless technologies, albeit with a limited communication range of approximately a hundred metres⁵⁰. In summary,

optical fibres and free-space links remain the primary choices for QKD quantum channels, each suited to different deployment scenarios and, consequently, facing distinct practical challenges.

Practical challenges of QKD

Application challenges arise from the gap between practical performance and expected demands, as well as the substantial trade-offs required to gain the benefits. The performance of QKD is prized for its key rate and practical security, but additional costs stem from the specialized requirements of quantum devices and channels.

Key rate

QKD supports key agreements for encryption. Once the key has been negotiated, the system then operates as classical encryption, ensuring compatibility with conventional devices. Explicitly, information-theoretical security can be achieved using a one-time pad, where the key must be as long as the data sequence. However, the key generation rate of practical QKD systems is inherently limited by the response rate, presenting challenges for encrypting large volumes of classical data transmission. Therefore, understanding and overcoming the constraints on QKD capacity are paramount.

Channel attenuation. Channel loss occurs when the photons flying in the quantum channel have a probability of disappearing, rendering them undetectable. This phenomenon can be described as the quantum states splitting into the environment. Despite the efforts of manufacturers to purify the glass-core fibre to minimize absorption, intrinsic inhomogeneities still cause Rayleigh scattering, leading to optical attenuation throughout the fibre. In particular, the low-loss fibre used in QKD experiments typically achieves attenuation as low as 0.16 dB km^{-1} (refs. 51,52), which is slightly lower than the standard telecommunication fibre attenuation of 0.2 dB km^{-1} . In the absence of a quantum repeater, the probability of photon detection will decay with the accumulation of channel loss⁵³, limiting the operational distance of QKD networks.

For point-to-point QKD over a pure-loss channel with a transmittance of η , the Pirandola–Laurenza–Ottaviani–Banchi (PLOB) bound⁵⁴ provides a tight limit on the secret key capacity. This bound demonstrates that the secret key rate per channel use is constrained to $-\log_2(1 - \eta)$, approximating a near linear scale of 1.44η at low transmittance (Fig. 2a). Hence, it is sometimes referred to as the linear bound. This bound is achievable in CV-QKD with the use of quantum memory. In practical scenarios operating without quantum memory, the secret key capacity of CV-QKD is reduced to 0.72η , which is lower than that typically achieved by DV-QKD. Specifically, the secret capacity of both BB84-like QKD with decoy states²⁵ and differential-phase-shift QKD with block-wise phase randomization⁵⁵ is proportional to η at low transmittance. Moreover, the reduced tolerance for channel loss in CV-QKD, combined with challenges in the post-processing procedure, leads to the more frequent application of DV-QKD in metropolitan QKD networks.

Consequently, developing new protocols⁵⁶ that can mitigate the limitations of secret key capacity imposed by channel loss and constructing quantum channels with lower attenuation – such as free-space links and hollow-core fibres⁵⁷ – are critical for establishing a global QKD network.

QKD signalling rate. When the secret key capacity per channel use is fixed, the key rate of a QKD system is typically proportional to the relative frequency of channel uses. However, this repetition rate is limited

by the speed of optical and electrical modulation and demodulation. Specifically, device bandwidth limitations can affect the accuracy of state preparation and measurement, whereas time jitter from the source, SNSPD and time tagger can lead to crosstalk between adjacent signals, all contributing to an increase in quantum bit error rate. Implementations of differential-phase-shift QKD have reached speeds up to 10 GHz in 2007 (ref. 58), as the issue of crosstalk is less severe in this context. Nonetheless, protocols such as coherent-one-way QKD¹⁴ have encountered challenges regarding security vulnerabilities under general attacks⁵⁹. Owing to the deterministic phase relation of all pulses, the secret key capacity of coherent-one-way QKD has been proven to scale as $O(\eta^2)$ ^{16,60}, indicating that such schemes are not suitable for long-distance deployments. For one-way BB84-like protocols, QKD systems have achieved repetition rates of up to 5 GHz for polarization encoding⁶¹ and 2.5 GHz for time-bin and phase encoding⁶².

To further increase the repetition rate, it would be crucial to enhance device performance⁶³, design ultrafast optical modulation schemes⁶⁴ or develop passive protocols⁶⁵. However, the key rate increases only linearly with the repetition rate, which has already approached a bottleneck at the gigahertz level. Moreover, further increasing the repetition rate introduces additional side-channel loopholes, such as correlations between pulses⁶⁶. Consequently, increasing the repetition rate is not a promising focus for the QKD community.

Detector saturation. Assuming low-loss channels and high repetition rates, the saturation of single-photon detectors in DV-QKD could still limit the key rate of the system⁶⁷. This saturation arises owing to the dead time, during which SPADs or SNSPDs become temporarily insensitive to incoming photons as they reset to their initial state for subsequent detection. To address this, multipixel SNSPDs have been developed⁶⁸ to maximize the key rate (Fig. 2b). When one pixel detects a photon and begins resetting, the remaining pixels are still active and ready to detect additional photons. The multipixel design not only shortens the dead time by reducing the length of each pixel but also reduces time jitter⁶⁹. This enhancement improves the saturation count rate of SNSPDs and subsequently increases the overall key rate to 110 Mb s^{-1} (ref. 70), approaching the throughput achievable in 4G networks.

Cost

When deploying a large-scale QKD network, it is crucial to strike a compelling balance between benefits and costs. Similar to many new technologies, the implementation of QKD networks remains expensive, often costing hundreds of times more than deploying classical networks of a similar scale.

Channel occupation. Establishing stable quantum links is fundamental for constructing QKD networks. However, extensive resources and close cooperation with governmental bodies are required. For instance, satellite links and other aerial vehicles are necessary, and even the ground stations are ideally situated on tall buildings or on plateau regions. For instance, a metropolitan entanglement-based free-space network⁷¹ can be established using a tall building as the central node. Fortunately, the presence of dense classical networks provides potential fibre link resources for QKD network deployment, making it a cost-efficient option to share fibres with classical communication. However, the coexistence of quantum signals with classical optical communications demands effective isolation of the quantum channel from noise⁷². Using a dedicated frequency band⁴⁹, along with high-isolation wavelength division multiplexing⁷³ devices

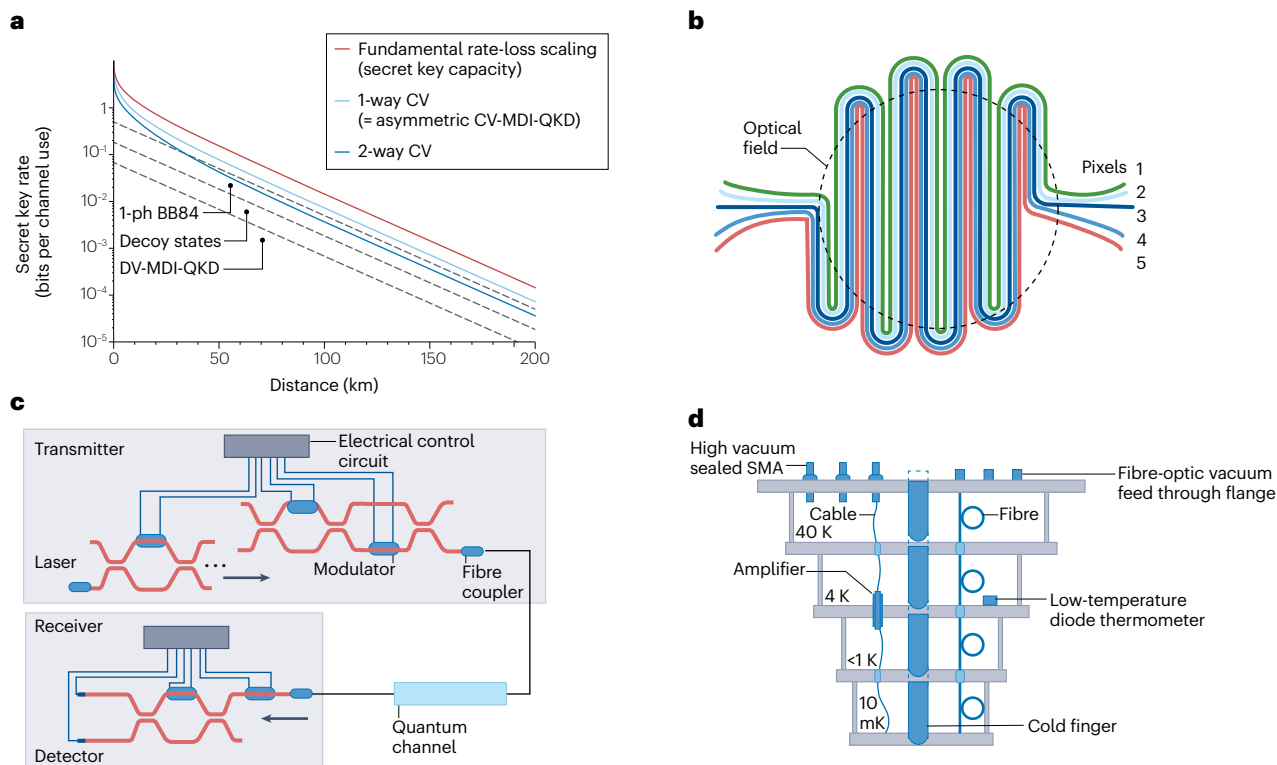


Fig. 2 | Fundamental limitations of quantum key distribution schemes. **a**, Simulation of ideal performances for different quantum key distribution (QKD) schemes⁵⁴, including the following abbreviations: continuous-variable (CV); discrete-variable (DV); measurement-device-independent (MDI); Bennett–Brassard 1984 protocol with single-photon sources (1-ph BB84). **b**, Schematic

of a superconducting nanowire single-photon detector (SNSPD) chip with multiple interleaved pixels. **c**, Schematic of on-chip QKD systems. **d**, Schematic of multistage cryogenic cooling system with Sub-Miniature version A (SMA) connectors. Part **a** is adapted from ref. 54, CC BY 4.0.

or optical filters⁷⁴, can help mitigate environmental disturbances. Even so, in-band noise caused primarily by Raman scattering⁷⁵ from high-power classical signals⁷⁶ must also be addressed⁷⁷ to ensure the efficient use of bandwidth resources.

However, certain devices commonly used in classical networks, such as reconfigurable optical add-drop multiplexers and optical communication repeaters⁷⁸, are not suitable for quantum signals. Instead, the quantum channel must bypass these devices to prevent the quantum signals from being disturbed. Furthermore, although classical optical communication can transmit data over hundreds of kilometres of fibre with the help of repeaters, quantum signals over such long distances become too weak relative to the background noise⁷⁴. Consequently, a ready-to-use integration of a QKD network into existing classical networks is not yet feasible. Designing the appropriate QKD network infrastructure⁷⁹ and coordinating with telecommunications companies are required before such integration can be realized.

Device form factor. For widespread applications of QKD networks, quantum devices are expected to achieve a similar level of miniaturization as those in classical networks. Dense integration of quantum devices can enhance functionality and enable the construction of scalable QKD systems. In fact, QKD systems require precise modulation of quantum states, and lithographic precision makes this achievable even in mass production. Moreover, integrated structures make components such as Mach–Zehnder interferometers robust

against environmental noise⁸⁰, reducing the need for feedback systems²⁹.

Much of the research on on-chip QKD is based on silicon and silica photonics⁸¹ owing to its mature manufacturing techniques. Modulation on silicon chips primarily relies on thermo-optic effects, whose slow response limits the choice of QKD protocols⁸². However, carrier depletion in silicon enables high-speed modulation for active QKD⁷⁰. Additionally, developments in on-chip lithium niobate⁸³ provide an alternative by facilitating easier electro-optic modulation.

The main advantages of chip-based QKD systems lie in achieving monolithic hybrid integration (Fig. 2c). For QKD transmitters, it is crucial to integrate the light source directly onto the chip⁸⁴, making materials such as InP⁸⁵ preferable for this purpose. For QKD receivers, integrating SNSPDs⁸⁶, SPADs⁸⁷ and homodyne detectors^{88,89} onto the chip is essential. However, practical issues such as stability and optical loss in coupling communication fibres to chips must be carefully considered⁹⁰. Solving these issues would enable the mass production of components for QKD networks and consequently reduce the cost of integration and worldwide commercial deployment.

Cooling requirements. Single-photon detectors operating in the telecommunication wavelength range require much lower temperatures than classical detectors to minimize noise. For example, SNSPDs achieve optimal performance only at temperatures much below half the critical temperature of their superconducting material – practically

around 2–4 K for Nb(Ti)N-based SNSPDs⁹¹. This temperature necessitates multistage cooling systems (Fig. 2d), such as Gifford–McMahon⁹¹ or Joule–Thomson cryocoolers⁹², which have considerable size, weight and power requirements when implemented at base stations within classical networks. Although there is ongoing research to develop SNSPDs from materials with higher critical temperatures⁹³, the fundamental operating principle of SNSPDs implies that the cost of cryogenic cooling systems cannot be entirely eliminated. Centralizing the SNSPD chips could reduce the number of cryogenic systems needed, crucially bringing down the cooling costs for the overall structure of QKD networks. Moreover, advances in cost-effective cryocoolers will have a crucial role in shaping the future infrastructure of QKD networks.

By contrast, SPADs can operate near or even at room temperature⁹⁴, although they generally suffer from lower detection efficiency and higher noise levels. A typical thermoelectric cooler provides adequate cooling for SPADs, avoiding the need for cryogenic systems and offering an economical and flexible option for equipping QKD receivers⁹. Consequently, in advanced and well-funded research groups, SNSPDs are commonly used for detecting photons in the telecommunication band. In more commercially oriented settings, however, SPADs remain the preferred choice. As global QKD transitions to commercial applications, reducing the cooling requirements may shift the cost–performance balance and ultimately reshape the motivation for building QKD networks.

Practical security

QKD promises to guard against all types of attacks encountered in quantum mechanics and information theory, thereby providing information-theoretic security for shared keys. However, despite the rigorous derivations in security analysis, it generally assumes that all user devices are perfectly modelled – an assumption that rarely holds in practice. Such discrepancies could lead to unexpected information leakage beyond the bounds set by the security analysis, a phenomenon known as side channels. These side channels involve highly sensitive quantum devices designed to ensure rigorous security. Therefore, addressing these side-channel vulnerabilities is crucial for the security of QKD applications.

Attack on devices. A practical QKD system may be vulnerable to various quantum attacks owing to device imperfections (Table 1).

In response to the challenges posed by these quantum attacks, effective countermeasures have been proposed. Countermeasures can be broadly categorized into two main types: suitable monitoring modules and machine-learning detection modules. In monitoring modules, leveraging suitable monitoring devices to track the physical parameters of pulses, such as intensity, phase and wavelength, guards against specific quantum attacks. However, these monitoring devices can introduce potential security vulnerabilities owing to their inherent imperfections and increase the overall complexity of QKD systems. In detection modules, machine-learning algorithms identify various patterns and features of quantum attacks to achieve universal attack detection in specific protocols. Although these machine-learning detection modules can handle most quantum attacks on CV-QKD protocols, they are unable to detect quantum attacks on other protocols. Therefore, it is essential to develop a universal countermeasure that can protect any practical QKD protocol from various quantum attacks. For a comprehensive overview of quantum attacks on QKD systems, readers can refer to the report ‘Implementation Attacks against QKD Systems’ commissioned by the German Federal Office for Information Security (BSI)⁹⁵.

Research on potential attack methods exploiting specific device loopholes has spurred the development of corresponding countermeasures. Although these collective efforts have clarified the issue of practical security, relying on additional devices to monitor existing ones remains a reactive rather than a forward-looking solution.

Measurement-device-independent QKD. In one-way quantum communication, attacks on the transmitter’s devices are relatively less concerning as they can be more easily monitored⁹⁶. However, isolators are not suitable for the receiver, and the measurement of quantum states requires the receiver’s devices to be highly efficient and sensitive. Consequently, many challenges in defending QKD systems against practical attacks are linked to their measurement devices. To address this issue, measurement-device-independent QKD (MDI-QKD) protocols⁹⁷, also known as side-channel-free QKD⁹⁸, were proposed. These protocols eliminate the need for specific assumptions about modelling of the measurement devices.

To elaborate a little further, in MDI-QKD protocols, two authorized users each generate quantum states, whereas a third party positioned

Table 1 | Typical quantum attacks against quantum key distribution

Quantum attack	Target device ^a	Protocol	Countermeasures	Refs.
Calibration attack	Local oscillator	Continuous-variable (CV)	Hidden-Markov-model-based recognition	156
Saturation attack	Homodyne detector	CV	Gaussian post-selection	157
Wavelength attack	Beam splitter	CV	Incorporating a simple wavelength filter	158
Laser-seeding attack	Source	Discrete-variable (DV)	Incorporating an external isolator	159
Source attack	Source	DV	Implementing phase randomization	160
Phase-remapping attack	Phase modulator	DV	Enhancing timing control and implementing phase randomization	161
Blinding attack	Homodyne detector Avalanche photodiode	CV DV	Using a sensitive p–i–n photodiode	162,163
Local oscillator –intensity attack	Local oscillator	CV	Monitoring the local oscillator intensity	164
Wavelength-switching attack	Acousto-optic modulator	DV	Well-calibrated modulator	165 ^a

^aThe specific devices under attack, as well as the corresponding countermeasure methods, differ depending on the attack schemes associated with each protocol.

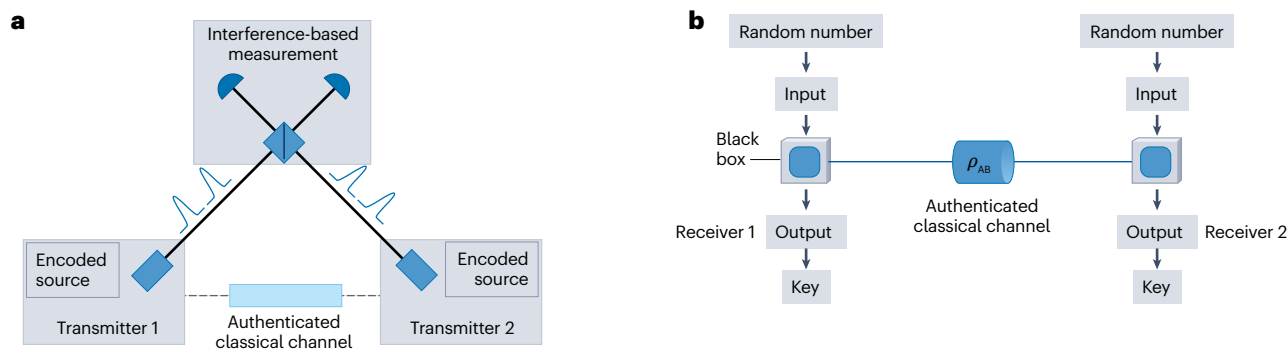


Fig. 3 | Advanced technologies in quantum key distribution developed to address practical challenges. **a**, Schematic of an interference-based quantum key distribution system. **b**, Schematic of device-independent quantum key distribution protocol. ρ_{AB} represents the entangled state distributed to the two receivers.

between them performs interference-based measurements (Fig. 3a). Although high-fidelity measurements are still required in MDI-QKD, they do not need to be trusted. The third party is treated similar to the quantum channel, with security verified through protocol procedures. It is worth noting that typical MDI-QKD protocols require two-photon interference with pre-selected measurements, resulting in roughly the same secret key capacity as point-to-point QKD, which scales as $O(\eta)$. Although MDI-QKD resolves security concerns related to measurement devices, it introduces additional untrusted nodes and necessitates precise synchronization to account for timing differences between the two channels. Moreover, practical setups with asymmetric node positions can decrease secret key capacity⁹⁹. As a result, MDI-QKD has not been as widely applied as point-to-point QKD for network establishment, awaiting the development of new protocols that surpass the PLOB bound⁵⁶.

Device-independent QKD. Device-independent QKD (DI-QKD) protocols¹⁰⁰ step further by eliminating the need for any specific assumptions about the modelling of all quantum devices, aiming for the ideal level of security promised by QKD¹⁰¹. In a DI scenario, authorized users treat local quantum devices as black boxes, focusing solely on their classical inputs and outputs rather than on their internal operations. As a result, any potential flaws in the quantum devices owing to intrinsic imperfections or malicious tampering can, in principle, be eliminated. Specifically, DI-QKD distributes the secure key through non-local correlation¹⁰⁰, with the parameter estimation resembling the process of verifying the violation of Bell inequality (Fig. 3b). However, a key challenge remains in addressing the detection loophole. In DI-QKD, only one of the detectors clicks, known as non-coincidence outcomes, must be accounted for rather than discarded during post-selection, similar to the requirements of the loophole-free Bell test¹⁰². This loophole makes photonic approaches¹⁰³ to DI-QKD challenging for long-distance applications, unless improved protocols¹⁰⁴ can relax the stringent detection efficiency requirements such as relying on a qubit amplifier¹⁰⁵ or the remote Bell test¹⁰⁴. By contrast, DI-QKD based on the establishment of distributed entanglement, such as systems using rubidium atoms¹⁰⁶ and trapped ions¹⁰⁷, can close the detection loophole through single-shot readout, offering a promising solution for DI-QKD networks.

However, DI-QKD still cannot conduct entirely loophole-free experiments. Unlike in a loophole-free Bell test, in which the locality loophole requires measurements to be separated to exclude

hidden-variable influences. This concern is not as critical in DI-QKD. The general settings of DI-QKD inherently close the locality loophole under the assumption that quantum theory is correct and that no unintended information, such as the input and output, leaks from the users' ends¹⁰⁶. Nevertheless, certain assumptions about the devices such as the perfect isolation¹⁰⁷ still persist, leaving potential loopholes. As DI-QKD becomes widely applied in networks, these issues could still limit the implementation reliability, reducing their competitiveness against classical networks. Consequently, DI-QKD appears to be bottlenecked by the technologies required for establishing distributed entanglement, as well as by the challenge of addressing the final unresolved aspect of practical security. As a result, the community's efforts and interests are increasingly directed towards extending the scale of QKD networks.

Long-haul QKD

Long-haul QKD serves as the backbone for establishing connectivity in global quantum networks. Since the 1990s, researchers have made substantial progress in this area (Fig. 4). In 1993, the first secure QKD link over 10 km was demonstrated using BB84 (ref. 108). This was followed by the plug-and-play system in 2002, which extended the distance to 67 km (ref. 109). The introduction of decoy-state methods^{25,26} enabled distances beyond 100 km by 2007 (ref. 110) and over 250 km by 2009 using ultra-low-loss fibres¹¹¹. By 2016, MDI-QKD achieved a record fibre-based QKD distance of 404 km (ref. 112). 2017 saw the first demonstrations of satellite-to-ground QKD⁴⁸ and intercontinental satellite-relayed networks¹¹³. In 2018, the twin-field QKD (TF-QKD) protocol was introduced, breaking the theoretical limit of point-to-point protocols⁵⁶. Using ultra-low-loss fibres, TF-QKD reached 509 km in 2020 (ref. 114) and surpassed 830 km in 2022 (ref. 115). In parallel, the mode-pairing QKD (MP-QKD) protocol was proposed in 2022 (refs. 116,117). In 2023, TF-QKD extended secure QKD transmission beyond 1,000 km, setting a new benchmark for practical long-distance quantum communication¹¹⁸. Despite this progress, QKD must sustain high key rates at such distances before it can be implemented in global networks.

Trusted relay

Optical loss increases with distance in quantum communication the same as in classical systems. To extend distance, the most direct approach is to add relays¹¹⁹. Although quantum repeaters offer a trust-free solution in principle¹²⁰, they remain costly¹²¹ and impractical for near-term

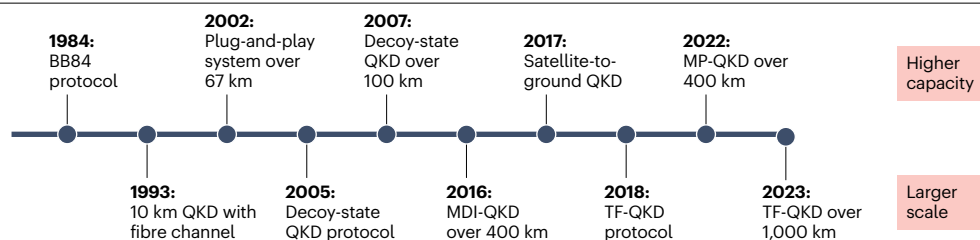


Fig. 4 | Timeline of long-haul quantum key distribution developments. 1984, Bennett–Brassard 1984 (BB84) protocol¹¹; 1993, 10 km quantum key distribution (QKD) with fibre channel¹⁰⁸; 2002, plug-and-play system over 67 km (ref. 109); 2005, decoy-state QKD protocol²⁵; 2007, decoy-state QKD over 100 km (ref. 110);

2016, measurement-device-independent (MDI)-QKD over 400 km (ref. 112); 2017, satellite-to-ground QKD⁴⁸; 2018, twin-field (TF)-QKD protocol⁵⁶; 2022, mode-paring (MP)-QKD protocol¹¹⁶; and 2023, TF-QKD over 1,000 km (ref. 118).

deployment. By contrast, trusted relays are often installed in physically secured suburban facilities, but substantial operational resources are required for maintaining real-time monitoring against eavesdropping.

The European project Secure Communication based on Quantum Cryptography (SECOQC)¹²² is an early example of using quantum relays to implement a quantum network. Subsequently, trusted relays have been used in many quantum networks (Fig. 5). How to deploy nodes as trusted relays in a multiuser quantum network to optimize the network's efficiency and the number of relays is also an important issue. There are many models and algorithms regarding this aspect¹²³. Another approach would be to eliminate trusted relays altogether. MDI-QKD⁹⁷ allows the use of untrusted measurement nodes, further reducing the need for trusted relays in quantum networks and thereby the security of quantum networks and mitigating the practical costs. The best approach may be to use both trusted and untrusted relays to maximize the efficiency of quantum networks in terms of cost and performance for the future development of a global quantum network¹²⁴.

Hybrid quantum-safe relay

PQC is widely considered as a compelling quantum-safe alternative to QKD. PQC can effectively mitigate security concerns associated with trusted relays in quantum networks, a concept referred to as secure relay¹²⁵ or hybrid quantum-safe cryptosystem¹²⁶. Combining PQC with quantum communication can further enhance security, with each method mutually reinforcing the other.

To secure trusted relays across quantum networks, one intrinsic solution involves transmitting ciphertext using quantum-secure direct communication^{127,128}. In this method, only information pre-encrypted by PQC is encoded onto quantum states and transmitted through quantum networks. This guarantees that, even if any network relay is compromised, the security of all encrypted messages remains quantum-safe. Alternatively, PQC can be integrated with the post-processing procedure of QKD for quantum-safe key derivation¹²⁶. Furthermore, PQC can have a vital role in public key generation and authentication processes¹²⁹, establishing a robust and reliable classical infrastructure to support quantum networks. PQC complements quantum communication by mitigating vulnerabilities of trusted relays, securing ciphertext transmission and strengthening key management and authentication, thereby establishing a robust hybrid framework for quantum networks.

Overcoming the bounds of repeater-less systems

The PLOB bound was initially considered as the definitive bound for all repeater-less point-to-point QKD systems^{53,54}. However, it was

subsequently recognized as inapplicable to interference-based QKD, which operates beyond the conventional point-to-point framework. In 2018, the TF-QKD protocol⁵⁶ allowed for effective events from single-photon responses to be directly used for key generation. This protocol is based on the principle of single-photon interference, in contrast to typical MDI-QKD protocols that rely on two-photon interference. Consequently, the key generation rate overcomes the PLOB bound and is rather proportional to the square root of the transmission rate. In 2022, the MP-QKD protocol, developed¹¹⁶ and concurrently proposed as asynchronous MDI-QKD¹¹⁷, exceeded the PLOB bound for key generation. The MP-QKD introduces a measure-then-pair strategy based on MDI-QKD, enhancing the utilization rate of response events. These protocols also allow the key rate to be proportional to the square root of the transmittance.

To compare the two methods, TF-QKD protocol imposes extremely stringent requirements on the wavelength matching between the two communicating parties' lasers, typically necessitating a frequency difference of less than 1 Hz (ref. 56). By contrast, conventional MDI-QKD and MP-QKD are considerably more tolerant, allowing a central wavelength mismatch of up to 125 MHz (refs. 29,130). Moreover, TF-QKD and MP-QKD have been validated in laboratory settings^{131–133} and field fibre-optic environments^{43,134,135}. However, despite surpassing the repeater-less bound, these single-interference-based protocols still do not meet the single-repeater bound of $-\log_2(1 - \sqrt{\eta})$ for end-to-end QKD¹³⁶. The trade-off between key rate and channel loss still exists. Consequently, exploring strategies¹³⁷ to surpass this new bound at the protocol level remains a crucial area for future research in global QKD.

Satellite QKD

It is anticipated that in the future, satellite-based quantum networks can better support global QKD. For instance, the European Quantum Communication Infrastructure (EuroQCI) project, involving all 27 EU Member States, will consist of a terrestrial segment based on fibre communication networks and a space segment utilizing satellites¹³⁸.

For a satellite-based quantum network, multiple satellites are required to complete seamless coverage of the system. Satellites can be typically divided into low-Earth orbit (LEO) satellites, medium-Earth orbit (MEO) satellites and geostationary-Earth orbit (GEO) satellites. The simplest satellite-based QKD network consists of an MEO satellite and three LEO satellites⁴⁴. LEO satellites have shorter orbital periods, allowing them to provide 24-h service, but they cover a smaller area so that more of them are needed for seamless coverage. By contrast, GEO satellites have longer orbital periods, meaning that the effective time for QKD within a day is shorter, but they cover a larger area. The benefits

and trade-offs of MEO lie between those of LEO and GEO, as its orbital altitude is intermediate between the two. Therefore, when building a satellite-based QKD network, LEO, MEO and GEO satellites are needed to achieve efficient QKD.

Satellites use free-space attenuation. Unlike exponential loss in fibres, free-space attenuation depends on atmospheric thickness, creating a ~ 20 dB loss over 1,000 km (ref. 44). Satellites thus could enable global QKD across countries and continents¹³⁹. In 2008, the feasibility of single-photon transmission was initially verified via satellite-to-ground links¹⁴⁰. Subsequently, in 2013, the feasibility of QKD under high-speed motion using an aircraft platform was demonstrated¹⁴¹. Quantum communication on a high-altitude balloon platform was also achieved¹⁴². As of 2025, the largest project is the Micius Quantum Science Experimental Satellite⁴⁴. Two ground stations connected by Micius, Xinglong and Nanshan, are 2,600 km apart. This communication distance was previously the longest achieved in single-relay QKD but has now been extended to 12,900 km (ref. 10).

Finally, the size and weight of satellite systems are also critical factors for successful operation¹¹³. More integrated systems offer smaller form factors for QKD systems and could be potential solutions for quantum satellite networks¹⁴³. In summary, satellite-based quantum networks provide a crucial pathway to extend QKD beyond the limits of optical fibres, enabling global-scale secure communication. Leveraging complementary LEO, MEO and GEO satellites together with integrated satellite systems will be key to building global quantum networks.

Fibre-based terrestrial QKD networks

As QKD evolves towards large-scale deployment, building multiuser, multinode quantum networks becomes inevitable. Practical implementation must consider diverse communication distances, asymmetric demands and interchannel crosstalk. For terrestrial links, optical fibres are the preferred medium owing to low loss, electromagnetic immunity and deployment readiness, despite demonstrations using free space and even underwater channels^{144,145}, where commercial fibres generally meet quantum communication requirements. Depending on the transmission scale, fibre-based QKD networks are typically classified as metropolitan or intercity.

Metropolitan areas. In metropolitan area QKD quantum networks, the distance of optical fibres is usually only a few kilometres to tens of kilometres but usually requires many nodes. For example, the quantum network in Hefei has 46 nodes¹⁴⁶. It is therefore necessary to design a reasonable topology to optimize the overall key generation efficiency. Meanwhile, various practical factors in node deployment must also be considered, such as the fact that some locations may not be geographically suitable for use as intermediate trusted relays. In metropolitan networks, MDI-QKD reduces system complexity by enabling detector sharing, whereas optimized protocol parameters compensate for channel asymmetry⁹⁹. One of the most straightforward ways to address an asymmetric path is to apply attenuation compensation to the less-attenuated segment, making the situation symmetrical. However, the parameters under asymmetric conditions have been

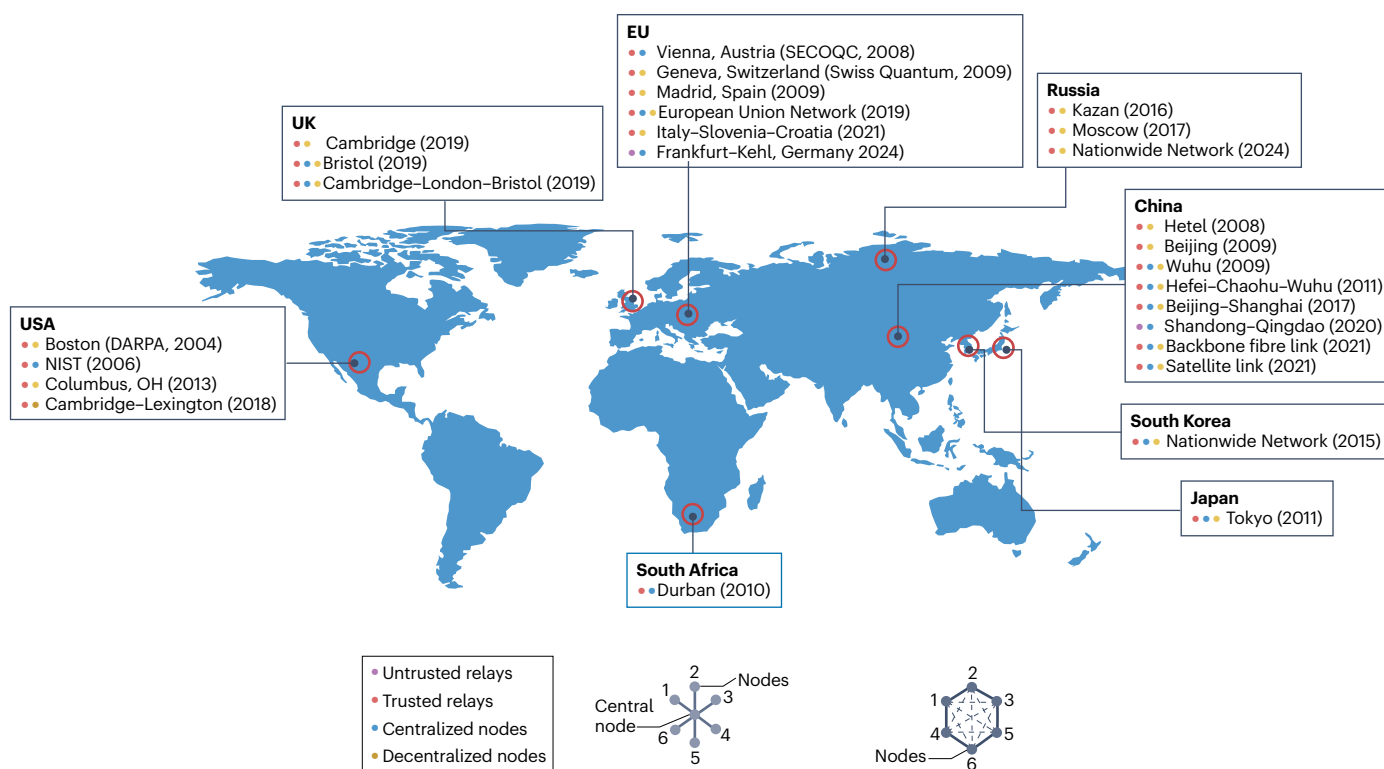


Fig. 5 | Main existing quantum key distribution networks worldwide. Representative quantum key distribution networks have been deployed worldwide, including national and metropolitan implementations in Europe^{9,138,166}, the USA⁷², China^{143,148,167}, Japan¹⁶⁸ and others⁷⁹. The architectures of existing quantum key

distribution networks may vary in design, but they are generally classified according to two main distinctions: trusted versus untrusted relays and centralized versus decentralized topologies.

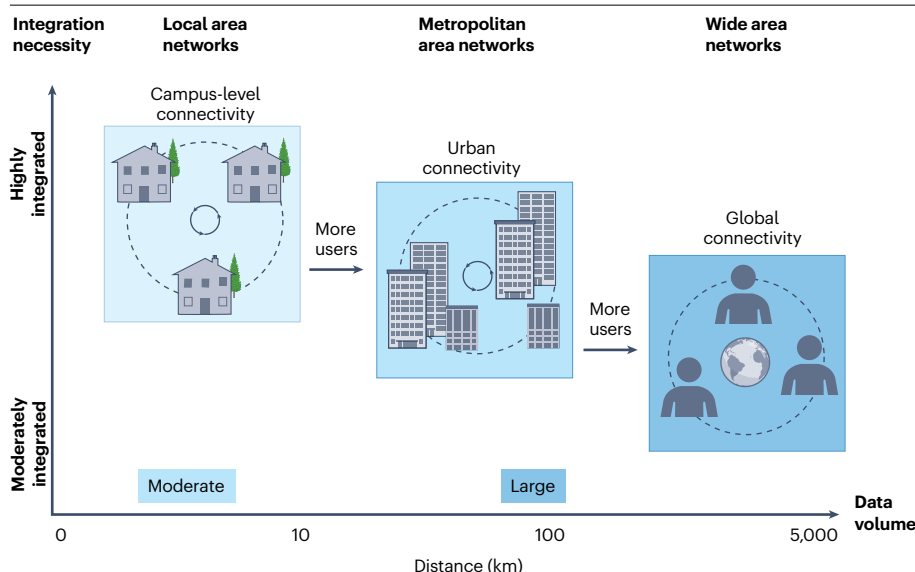


Fig. 6 | An envisioned layout for a global quantum key distribution network. Range definitions of local area networks, metropolitan area networks and wide area networks are approximated from classical networking. Local area networks generally require a higher level of user system integration to meet user demands, whereas the integration requirements decrease for metropolitan and wide area networks.

optimized to achieve a higher coding rate without needing attenuation compensation, even surpassing the results obtained with added attenuation¹⁴⁷.

Optical switching and secure key management in practical metropolitan quantum networks are also crucial¹⁴⁸ for optimizing the end-to-end key rate of QKD network. The switching can be categorized into two types: matrix switching and fully connected switching. Matrix switching does not allow any two users to connect freely, whereas fully connected switching enables users connected to the optical switch to establish connections between two end points.

Overall, metropolitan quantum networks must balance topology design, protocol optimization and switching strategies to maximize key generation efficiency. Addressing channel asymmetry and enabling flexible interconnections are crucial steps towards building practical large-scale urban QKD infrastructures.

Intercity areas. Similar to metropolitan quantum networks, the setup of trusted relays in intercity networks also has to consider distance, fibre environment and cost to ensure their efficient operation. Intercity quantum networks typically span hundreds of kilometres, where the effects of photon attenuation, dispersion and scattering in optical fibres dominate⁴³. Several countries, including the USA, Europe, China and others, have already begun establishing intercity quantum networks (Fig. 5).

One challenge in establishing intercity quantum networks is that long-distance fibres also contain multiple splicing points, introducing reflection and interchannel crosstalk^{134,135}. Quantum information is typically transmitted alongside classical optical communication⁷⁶, either in different optical cables or even in different fibres within the same cable. This often requires accounting for the various interferences introduced by classical optical signals and using methods to reduce them. Most protocols have key rates scaling with the square of transmittance, making even small improvements in loss crucial for enhancing overall performance. Moreover, some classical synchronization signals often need to be amplified over intercity distances using optical amplifiers¹³⁴, making it even more crucial to address the crosstalk between the synchronization signals and optical quantum

signals. Quantum noise also greatly increases over large intercity distances, ultimately degrading signal-to-noise ratio and reducing the achievable secure key rate.

However, at intercity distances up to 200 km apart, the TF-QKD and MP-QKD protocols have demonstrated advantages regarding key rate and security. By exploiting single-photon interference and mode-pairing strategies, these protocols achieve a more favourable $\sqrt{\eta}$ scaling, surpass the PLOB bound and offer improved tolerance to channel loss and noise. These features are crucial for extending QKD from metropolitan links to practical intercity backbones, where both distance and stability are critical. The diversity of implementation methods¹⁴⁹ for TF-QKD and the simplicity of MP-QKD experimental setups give these two protocols even a broader application appeal. Therefore, TF-QKD and MP-QKD protocols are likely to become important protocols for future intercity quantum networks.

In summary, intercity quantum networks face notable challenges from loss, noise and classical crosstalk, making protocol and infrastructure optimization essential. TF-QKD and MP-QKD stand out as promising solutions, combining practical feasibility with enhanced key rates and security, and are therefore strong candidates for the backbone of future intercity quantum networks.

Outlook

To extend communication distances, the core challenge lies in capturing more of the encoded information carried by photons. Beyond the aforementioned approaches to increase the key rate, the use of quantum repeaters can enhance the key rate up to the N -repeater bound¹³⁶ of $-\log_2(1 - N+1/\eta)$. Although still far from realization, the establishment of quantum repeater networks⁸ could address major performance challenges, including key rate limitations and security loopholes. For example, long-distance entanglement can be established by combining short-distance entanglement generation with entanglement swapping¹²⁰, which also supports the implementation of DI-QKD. Regarding security, fully passive QKD protocols have also been proposed to eliminate side-channel vulnerabilities introduced by active modulation^{65,150} beyond what is possible through MDI-QKD and DI-QKD. High-dimensional QKD¹⁵¹ encodes quantum states in a Hilbert space with dimension $d > 2$

to improve noise tolerance. However, although the information density per mode $\log_2(d)/d$ can be slightly increased when $d = 3$, it tends to decrease for larger d , as higher-dimensional systems typically require more detectors. The measure-then-pair strategy introduced in MP-QKD offers an alternative route to enhance information density through techniques such as wavelength-division multiplexing¹⁵².

As a quantum counterpart to the internet¹⁵³, the global QKD network will consist of multiple layers, each tailored to specific application requirements. A potential layout for a future global QKD network considers factors such as transmission distance, user numbers, integration needs and data transmission volume across various application scenarios (Fig. 6). The challenge lies in the difficulty of quantitatively evaluating the practical security for the entire setup¹⁵⁴. Despite their paramount importance, security issues in QKD projects tend to receive less attention from funders and contractors, and key rates and costs are often prioritized instead. However, it is important to note that, compared with classical cryptography such as PQC, the only unique advantage of QKD lies in its unconditional security, which comes at the expense of capacity and cost. The motivation for establishing a global QKD network would be weakened if the strict requirements on security were to be deprioritized.

Additionally, different vendors and protocols often adopt distinct implementations and encoding schemes. Standardized interfaces and comprehensive testing frameworks, such as those published by the International Organization for Standardization and the International Electrotechnical Commission¹⁵⁵, are essential for enabling multivendor deployment, facilitating cross-border QKD operations and ultimately establishing a globally integrated, quantum-secure communication infrastructure.

If all of these challenges are addressed, global QKD promises to establish a worldwide, information-theoretically secure connection among all authorized users, offering practical capacity and cost-effectiveness.

Published online: 12 November 2025

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Acknowledgements

This work is supported by ASTAR (M21K2c0116 and M24M8b0004), Singapore National Research Foundation (NRF-CRP22-2019-0004, NRF2023-ITC004-001, NRF-CRP30-2023-0003, NRF-CRP31-0001 and NRF-MSG-2023-0002) and Singapore Ministry of Education Tier 2 Grant (MOE-T2EP50222-0018). The authors acknowledge the support of Dieter Schwarz Stiftung gGmbH for the QUASAR project. L.H. acknowledges the financial support of the following Engineering and Physical Sciences Research Council (EPSRC) projects: Platform for Driving Ultimate Connectivity (TITAN) under Grant EP/X04047X/1; Robust and Reliable Quantum Computing (RoarQ), EP/W032635/1; India-UK Intelligent Spectrum Innovation ICON UKRI-1859; PerCom, EP/X012301/1; EP/X01228X/1 and EP/Y037243/1.

Author contributions

H. Zhang and H. Zhu researched data for the article. All authors contributed substantially to discussion of the content and wrote the article. H. Zhang, H. Zhu, L.H. and W.G. reviewed and edited the manuscript before submission.

Competing interests

The authors declare no competing interests.

Additional information

Peer review information *Nature Reviews Electrical Engineering* thanks the anonymous reviewer(s) for their contribution to the peer review of this work.

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